Pendant Leaves on the Outer Orbit of a Bowtie: Double Symmetrization and Fractional Triangle Bias

Abstract

Let B be the bowtie graph, the vertex amalgam of two triangles. Its four noncentral vertices form one automorphism orbit. We attach h pendant leaves arbitrarily among those four vertices and prove the chromatic-polynomial graphon bound for $1 \le h \le 6$ and every $p \ge 1/2$. Averaging over the outer orbit gives degree exponent $h/4$ at each outer vertex. Conditioning on the shared center then factorizes the density as the integral of a square. Cauchy–Schwarz and a second symmetrization inside a triangle produce the uniform fractional weight $d^{h/6}$, which is controlled by a biased-measure Hölder argument and the accepted triangle bound. The case $h = 1$ proves the formerly open NetworkX Atlas graph 119.

1 Bowtie family and exact target

Let (Ω, μ) be an arbitrary probability space and let $W\colon \Omega^2 \to [0,1]$ be symmetric and measurable. Write

$$p = \int_{\Omega^2} W(x,y)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y), \qquad d(x) = \int_{\Omega} W(x,y)\,\mathrm{d}\mu(y).$$

Let B have center o and outer vertices a, b, c, d, with triangles oab and ocd. The automorphism group fixes o and is transitive on $\{a, b, c, d\}$: it may swap the two triangles and independently swap the endpoints within either outer edge.

For nonnegative integers n_a, n_b, n_c, n_d, put

$$h = n_a + n_b + n_c + n_d$$

and let $B(n_a, n_b, n_c, n_d)$ be obtained by attaching n_v private pendant leaves to outer vertex v. We assume

$$1 \le h \le 6.$$

The bowtie is the vertex amalgam of two triangles, so

$$\chi_B(x) = \frac{\chi_{K_3}(x)^2}{x} = x(x-1)^2(x-2)^2.$$

Each leaf supplies a further factor $x - 1$. Therefore

$$\chi_{B(\boldsymbol{n})}(x) = x(x-1)^{h+2}(x-2)^2, \qquad \chi(B(\boldsymbol{n})) = 3. \tag{1}$$

If

$$A_3(p) := p(2p-1),$$

then direct substitution in (1) gives

$$\Phi_{B(\boldsymbol{n})}(p) = p^h A_3(p)^2 = p^{h+2}(2p-1)^2. \tag{2}$$

This includes the polynomial continuation at $p = 1$; at the required threshold $p = 1/2$, it vanishes.

We use the accepted triangle bound in the form

$$t(K_3, V) \geq A_3(s) = s(2s - 1) \tag{3}$$

for every graphon V of edge density $s \geq 1/2$, on an arbitrary probability space.

2 Fractional triangle bias

Lemma 2.1 (Fractionally degree-weighted edge)**.** *For $0 < \beta \leq 1$, define*

$$N_\beta = \int_{\Omega^2} W(x, y) d(x)^\beta d(y)^\beta \, \mathrm{d}\mu(x) \, \mathrm{d}\mu(y).$$

Then

$$N_\beta \geq p^{1+2\beta}.$$

Proof. On $\{d(x) = 0\}$, define $W(x, y)/d(x)$ to be zero. Fubini implies that $W(x, y) = 0$ for almost every y there. Put

$$\theta = \frac{1}{1 + 2\beta}, \qquad \eta = \frac{\beta}{1 + 2\beta}.$$

Then $\theta + 2\eta = 1$, and weighted Hölder applied to

$$W = \left(W d(x)^\beta d(y)^\beta\right)^\theta \left(\frac{W}{d(x)}\right)^\eta \left(\frac{W}{d(y)}\right)^\eta$$

gives

$$p \leq N_\beta^\theta \left(\int \frac{W}{d(x)} \, \mathrm{d}\mu^2\right)^\eta \left(\int \frac{W}{d(y)} \, \mathrm{d}\mu^2\right)^\eta.$$

The quotient integrals are at most one, proving the claim. □

Lemma 2.2 (Triangle rescaling)**.** *Let $0 < h \leq 6$, let $p \in (1/2, 1]$, and suppose $0 < z \leq p^{h/2}$. Put*

$$s_0 = p \left(\frac{p^{h/2}}{z}\right)^{2/3}.$$

If $s_0 \leq 1$, then

$$z A_3(s_0) \geq p^{h/2} A_3(p).$$

Proof. For $s \in [1/2, 1]$, set

$$\psi(s) = \frac{A_3(s)}{s^{3/2}} = 2\sqrt{s} - \frac{1}{\sqrt{s}}.$$

Then

$$\psi'(s) = \frac{1}{\sqrt{s}} + \frac{1}{2s^{3/2}} > 0.$$

The definition of s_0 is equivalent to

$$z = p^{h/2} \left(\frac{p}{s_0}\right)^{3/2}.$$

Since $s_0 \geq p$, monotonicity gives

$$z A_3(s_0) = p^{(h+3)/2} \psi(s_0) \geq p^{(h+3)/2} \psi(p) = p^{h/2} A_3(p).$$

□

The function $A_3(s) = s(2s - 1)$ is nondecreasing on $[1/2, 1]$.

3 The outer-leaf theorem

Theorem 3.1 (Leaves on the bowtie's outer orbit)**.** *Let* $1 \le h \le 6$ *and let* $n_a + n_b + n_c + n_d = h$. *Every graphon* W *of edge density* $p \in [1/2, 1]$ *satisfies*

$$\boxed{t\big(B(n_a, n_b, n_c, n_d), W\big) \ge p^h A_3(p)^2 = \Phi_{B(\boldsymbol{n})}(p).}$$

Proof. The threshold $p = 1/2$ follows from nonnegativity, and at $p = 1$ both sides equal one. Assume $p \in (1/2, 1)$.

Integrating the leaves first gives the bowtie core integrand multiplied by

$$d(x_a)^{n_a} d(x_b)^{n_b} d(x_c)^{n_c} d(x_d)^{n_d}.$$

Average over the automorphism group of the bowtie. Its action is transitive on the four outer vertices, so each exponent has group average $h/4$. The arithmetic–geometric mean inequality therefore gives the lower bound in which every outer variable has exponent

$$\alpha = \frac{h}{4}.$$

For a fixed image x of the center, define

$$R_\alpha(x) = \int_{\Omega^2} W(x, y) W(x, z) W(y, z) d(y)^\alpha d(z)^\alpha \, \mathrm{d}\mu(y) \, \mathrm{d}\mu(z).$$

The two outer edges ab and cd use disjoint variables. Hence the symmetrized lower bound factorizes exactly as

$$t(B(\boldsymbol{n}), W) \ge \int_\Omega R_\alpha(x)^2 \, \mathrm{d}\mu(x) \ge \left(\int_\Omega R_\alpha(x) \, \mathrm{d}\mu(x) \right)^2, \tag{4}$$

where the second inequality is Cauchy–Schwarz.

The last integral is a triangle density with weight d^α at two of its three vertices. By triangle symmetry, the three choices of unweighted vertex give equal integrals. Average those three representations and apply the arithmetic–geometric mean inequality again. Every triangle vertex then has exponent

$$\beta = \frac{2\alpha}{3} = \frac{h}{6} \in (0, 1].$$

Consequently

$$\int R_\alpha \ge \int_{\Omega^3} W(x, y) W(y, z) W(z, x) d(x)^\beta d(y)^\beta d(z)^\beta \, \mathrm{d}\mu^3. \tag{5}$$

Put

$$M = \int_\Omega d(x)^\beta \, \mathrm{d}\mu(x), \qquad z = M^3, \qquad \mathrm{d}\nu(x) = \frac{d(x)^\beta}{M} \, \mathrm{d}\mu(x).$$

Here $M > 0$ because $p > 0$. The right side of (5) is $z \, t(K_3, W; \nu)$. The edge density on (Ω, ν) is

$$s = \frac{N_\beta}{M^2}.$$

Concavity of u^β and Lemma 2.1 give

$$0 < z \le p^{3\beta} = p^{h/2},$$

and

$$s \geq \frac{p^{1+2\beta}}{M^2} = p\left(\frac{p^{h/2}}{z}\right)^{2/3} =: s_0.$$

Since $s \leq 1$, also $s_0 \leq 1$; moreover $s \geq s_0 \geq p > 1/2$.

Apply the accepted triangle bound (3), its monotonicity, and Lemma 2.2:

$$\int R_\alpha \geq zA_3(s) \geq zA_3(s_0) \geq p^{h/2}A_3(p).$$

Substitution in (4) gives

$$t(B(\boldsymbol{n}), W) \geq p^h A_3(p)^2,$$

which is (2). □

Every balanced complete ℓ-partite graphon, $\ell \geq 2$, realizes equality. Its degree is constant, the rooted weighted-triangle function is constant by class symmetry, and the accepted triangle bound is sharp. At $\ell = 2$, both sides vanish.

4 Atlas 119, formalization audit, and limitations

Corollary 4.1 (Atlas 119)**.** *NetworkX Atlas graph* 119*, with graph6 code* `EtoO`*, is positive. It has order six, size seven, chromatic number three, and*

$$\chi_{F_{119}}(x) = x(x-1)^3(x-2)^2.$$

Every graphon of edge density $p \in [1/2, 1]$ satisfies

$$t(F_{119}, W) \geq p^3(2p-1)^2 = \Phi_{F_{119}}(p).$$

Proof. Its NetworkX edge list is

$$01,\ 02,\ 03,\ 04,\ 14,\ 23,\ 35.$$

The triangles 014 and 023 share center 0, and vertex 5 is a leaf at the outer vertex 3. Thus Theorem 3.1 applies with $h = 1$. □

All uses of Tonelli–Fubini concern bounded nonnegative measurable integrands. The proof explicitly handles $\{d = 0\}$ and proves the biased normalization positive before dividing. It uses ordinary homomorphism densities and assumes no regularity, finite-step model, minimum degree, or injectivity. The only accepted density input is the triangle theorem with exactly matching edge-density range.

The leaves must attach only to the four-vertex outer orbit. The shared center carries no leaf in this theorem. The bound $h \leq 6$ is the exact limit of this proof, because the second symmetrization produces exponent $h/6$ and the needed concavity reverses beyond one. The result does not cover arbitrary leaves on a vertex-amalgam, tails of length two, or a general mixed-chordal graph.

The graph identities, both symmetrized exponents, scalar derivative, and Atlas identification were reconstructed in exact rational arithmetic. For formal verification, the new pieces are the outer-orbit arithmetic–geometric symmetrization, conditional square factorization, Cauchy–Schwarz step, and the second triangle symmetrization. The graph-specific module awaiting inspection is `lean/Taeyoung/Examples/Graph119.lean`.

5 What the Lean formalization actually proves

The machine-checked development is `lean/Taeyoung/Methods/BowtieLeaf.lean`, ending at `satisfiesLowerBound`

Scope: $h = 1$ only. Theorem 3.1 is *not* formalized for $1 \le h \le 6$. What is proved is Corollary 4.1, on all of $p \in [1/2, 1]$, and it is the only member of the family inside the six-vertex catalogue: the bowtie already has five vertices, so $h = 1$. The two symmetrisation exponents are therefore $\alpha = 1/4$ and $\beta = 1/6$, both *reciprocal integers*, which is what lets the pre-existing `Methods/OddLeaf/Bias.lean` apply unchanged at $m = 6$: it builds $d^{1/m}$, $M = \int d^{1/m}$, the tilted measure ν and the reread kernel for every $m \ge 1$, and both $(d^{1/m})^m = d$ and $M^m \le p$ stay $\mathbb{N}$-powers. A general h would need $d^{h/6}$ with $h \nmid 6$, hence genuine `Real.rpow` arithmetic in those two places.

Lemma 2.1 was already available. It is the same statement as `odd_cycle_one_leaf.tex` Lemma 2.1 at $\beta = 1/m$, and is formalized once, for every m, as `OddLeaf.pow_le_pow_rootEdge`:

$$p^{m+2} \le N^m.$$

The polynomial form is deliberate — no real power of p appears in any statement. Atlas 119 uses the $m = 6$ instance `OddLeaf.pow_eight_le_pow_rootEdge`, $p^8 \le N^6$.

The departure: the rescaling carries no derivative and no square root. Lemma 2.2 is proved here by differentiating $\psi(s) = A_3(s)s^{-3/2} = 2\sqrt{s} - 1/\sqrt{s}$, and at $h = 1$ its statement contains $p^{h/2} = \sqrt{p}$. The formalization forms no square root at all. The observation is that the quantity the proof actually feeds into (4) is the *square* $(\int R_\alpha)^2$, so with $K = M^6 = z^2$ the whole rescaling can be stated squared,

$$p\,A_3(p)^2 \le K\,A_3(s)^2 \qquad (\texttt{BowtieLeaf.transfer}),$$

and its hypotheses are the two polynomial facts $K \le p$ (Jensen, `OddLeaf.pow_rootMean_le`) and $p^8 \le K^2 s^6$ (the fractional Hölder, through $N = M^2 s$). Clearing s^3 and dividing by $p^2 s^2$ reduces it to

$$p(2s-1)^2 - s(2p-1)^2 = (s-p)(4ps-1) \ge 0 \qquad \left(\tfrac{1}{2} \le p \le s\right), \tag{6}$$

a single `ring` factorisation — the same factor as the rescaling in `Methods/CliqueDist/Diamond.lean`. The intermediate s_0 never appears, and neither does the hypothesis $s_0 \le 1$; the inequality $p \le s$, which the note gets from $z \le p^{h/2}$ and the definition of s_0, is obtained instead from $p^8 \le K^2 s^6 \le p^2 s^6$ (`BowtieLeaf.le_biased`). The monotonicity of A_3 on $[1/2, 1]$, stated after Lemma 2.2, is not formalized and is not needed.

Both symmetrisations are measure-preserving permutations. "Average over the automorphism group of the bowtie" and "by triangle symmetry, the three choices of unweighted vertex give equal integrals" are discharged by `Methods/Peeling.integral_assignment_perm`, a general lemma extracting the measure-preserving half of `homDensity_iso` so that it applies to an integrand which is *not* a graph weight. For the outer orbit the three nontrivial automorphisms are given explicitly (`aut2`, `aut3`, `aut4`, each checked by `decide`); for the triangle they are the two rotations. Only transitivity on $\{a, b, c, d\}$ is used, never the full group. The two arithmetic–geometric mean steps are Mathlib's `Real.geom_mean_le_arith_mean_weighted` at four equal weights $1/4$ and at three equal weights $1/3$; the second one carries the exponent bookkeeping $\left(d^{1/4} d^{1/4}\right)^{1/3} \rightsquigarrow d^{1/6}$ (`BowtieLeaf.prod_pair_rootDegree`).

The factorisation (4) **is an identity, not an estimate.** `BowtieLeaf.integral_outer_factor` proves $\int_{\Omega^5} = \int_\Omega R_\alpha^2$ exactly, by peeling the two triangles into disjoint variable blocks; the Cauchy–Schwarz step $\int R^2 \geq (\int R)^2$ is separate (`BowtieLeaf.sq_integral_pairTri_le`, the project's `BookTail.sq_integral_mul_le` against the constant 1). The accepted triangle input (3) is `K4Tail.goodman`, applied on (Ω, ν) with no hypothesis on the biased density.